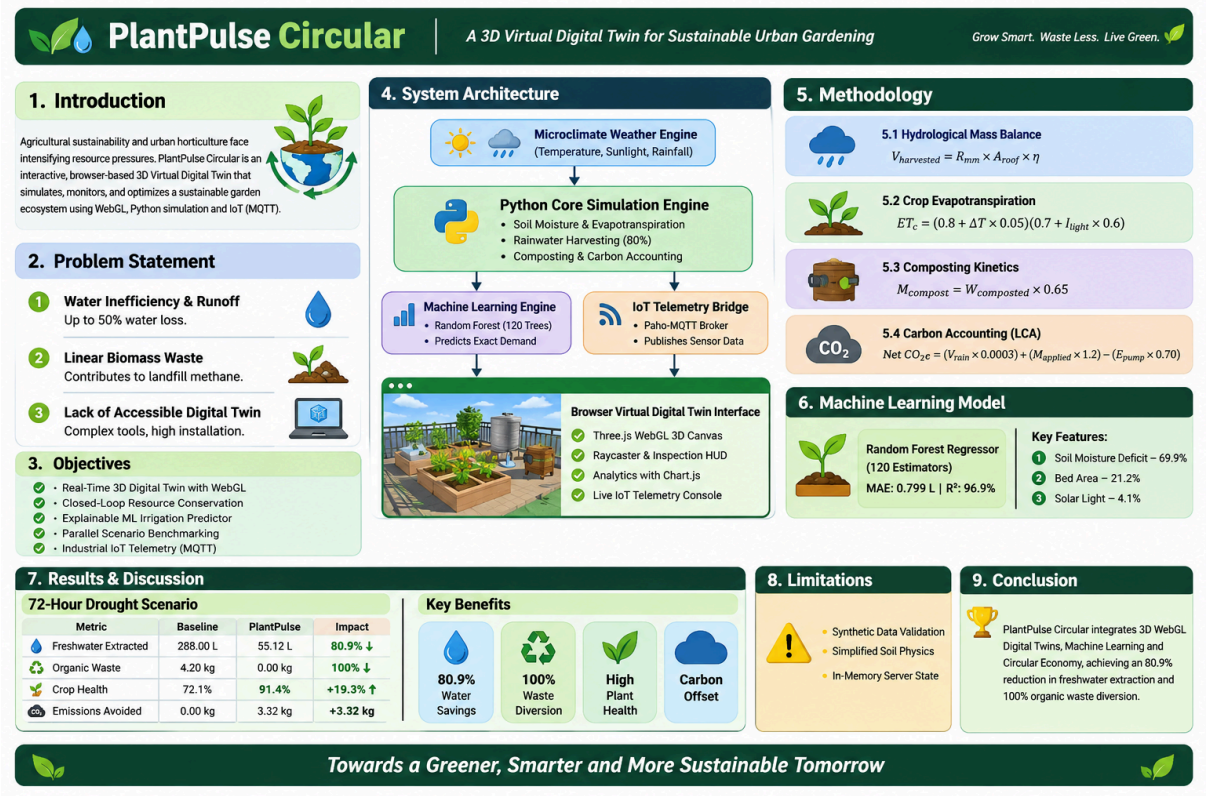
PlantPulse Circular - A 3D Virtual Digital Twin for Sustainable Urban Gardening

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1. Introduction

Agricultural sustainability and urban horticulture face increasing resource pressures due to freshwater scarcity, climate extremes, and soil nutrient depletion. Traditional open-loop cultivation practices often rely on manual observation or fixed-schedule irrigation systems. These methods may consume excessive municipal water while generating organic waste that is discarded in landfills.

PlantPulse Circular is an interactive, browser-based 3D Virtual Digital Twin designed to simulate, monitor, and optimize a sustainable urban garden ecosystem. Developed using WebGL, Three.js, a Python-based simulation engine, and an IoT messaging layer using MQTT, the system models a closed-loop micro-circular economy.

The proposed system integrates:

- Roof rainwater harvesting and automated cistern storage.
- Machine Learning-based irrigation demand prediction using a Random Forest Regressor.
- Organic pruning biomass conversion into compost through an aerobic bioreactor.

- Lifecycle carbon accounting and municipal utility savings estimation.
- Interactive 3D visualization and real-time environmental monitoring.

The system aims to demonstrate how digital twin technology, predictive intelligence, and circular resource management can improve the sustainability of urban gardening.

2. Problem Statement

Urban and domestic gardens experience three major inefficiencies.

2.1 Water Inefficiency and Runoff

Conventional fixed-schedule irrigation systems water crops uniformly, regardless of current soil moisture or expected rainfall. This can result in excessive evaporation, surface runoff, and nutrient leaching, reducing water-use efficiency.

2.2 Linear Biomass Waste

Garden pruning waste is frequently discarded as municipal solid waste. Its decomposition in landfills can contribute to methane emissions, while gardeners continue purchasing synthetic fertilizers to replenish soil nutrients.

2.3 Lack of Accessible Digital Twin Tools

Existing digital twin platforms often require specialized software, high-performance hardware, and advanced 3D programming skills. These requirements make them less accessible for small-scale urban farming, educational applications, and lightweight research prototypes.

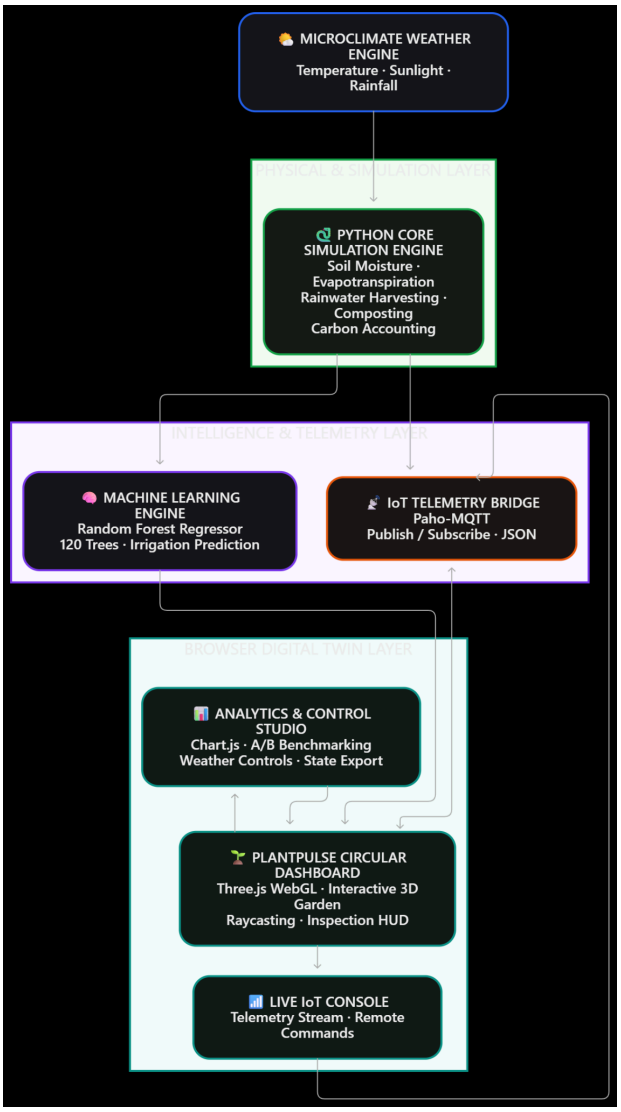
3. Objectives

The primary objectives of PlantPulse Circular are:

1. Develop a real-time, browser-based 3D Digital Twin using WebGL and Three.js.
2. Implement closed-loop resource conservation through rainwater harvesting, intelligent irrigation, and organic waste recycling.
3. Develop an explainable Machine Learning irrigation predictor targeting a Mean Absolute Error (MAE) below 1.0 L.
4. Enable parallel scenario benchmarking between conventional and circular gardening systems under identical weather conditions.
5. Establish an IoT telemetry architecture using MQTT for remote monitoring and simulated actuator control.

4. System Architecture

The PlantPulse Circular system is organized into three interconnected layers.



4.1 Physical and Simulation Layer

The simulation engine is implemented in backend/simulation.py. It models soil moisture, plant health, rainwater storage, irrigation demand, compost temperature, and carbon savings.

The simulation advances in deterministic six-hour intervals.

4.2 Intelligence and Telemetry Layer

The intelligence layer includes:

- backend/ml.py: Random Forest irrigation demand prediction.
- backend/mqtt_manager.py: MQTT telemetry publishing and command handling.

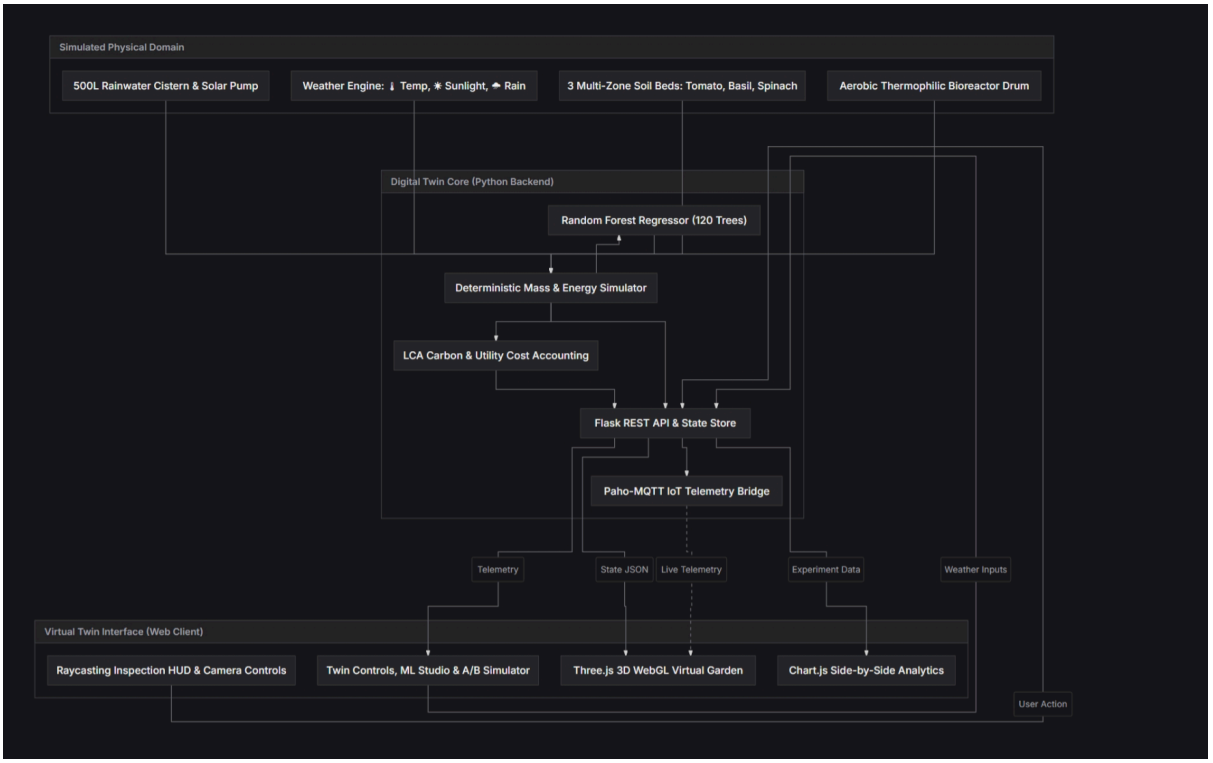
Structured JSON packets are published using topics such as plantpulse/telemetry/#.

4.3 Browser Digital Twin Layer

The visualization layer consists of:

- dashboard/twin3d.js: Three.js-based 3D garden visualization.
- dashboard/app.js: Dashboard controls, analytics, and interaction management.

The browser interface includes a WebGL viewport, interactive object inspection, comparative analytics, and an IoT telemetry console.



5. Methodology

5.1 Hydrological Mass Balance

Rainwater harvesting is calculated using rainfall intensity, roof area, and runoff efficiency.

Given:

- Roof area: 12 m²
- Runoff efficiency: 0.80
- Rainfall intensity: Rmm

The harvested rainwater volume is:

$V_{harvested} = R_{mm} \times A_{roof} \times \eta$

The cistern volume is updated using:

$$V_{tank}(t+1) = \min(V_{max}, V_{tank}(t) + V_{harvested}(t) - V_{rain_used}(t))$$

This equation represents the accumulation and utilization of harvested rainwater while ensuring that the tank capacity is not exceeded.

5.2 Crop Evapotranspiration and Deficit Dynamics

Crop water loss is estimated using ambient temperature and solar irradiance.

$$ET_c = (0.8 + \max(0, T_{air} - 20) \times 0.05) \times (0.7 + I_{light} \times 0.6)$$

The irrigation demand for each crop bed is calculated from its soil moisture deficit.

$$Demand = \max(0, 58 - \theta_{soil}) \times A_{bed} \times 0.12 \times \max(0.65, 1 + (T_{air} - 25) \times 0.015) \times (0.7 + I_{light} \times 0.6) \times K_c$$

Where:

- θ_{soil} represents soil moisture percentage.
- The target soil moisture is 58%.
- A_{bed} represents the bed surface area.
- I_{light} represents solar irradiance in the range [0, 1].
- K_c represents the crop coefficient.

Crop coefficients are:

Crop	Crop Coefficient
Tomato	1.05
Basil	0.75
Spinach	0.90

5.3 Biomass and Composting Kinetics

Pruning biomass of 0.35 kg per simulation interval is directed into an aerobic composting tumbler.

The biomass undergoes three maturity stages over 18 simulated hours, with a humus yield of 65%.

$$M_{compost} = W_{composted} \times 0.65$$

The compost pile temperature is modeled as:

$$T_{pile} = \min(58°C, 35°C + N_{batches} \times 6.5°C + T_{air} \times 0.25)$$

This model represents the increase in compost temperature due to microbial activity during decomposition.

5.4 Carbon Accounting

The lifecycle carbon accounting model estimates avoided emissions from municipal water pumping and synthetic fertilizer substitution.

$$\text{Net Avoided Emissions} = (V_{rain_used} \times 0.0003) + (M_{applied} \times 1.2) - (E_{pump} \times 0.70)$$

Where:

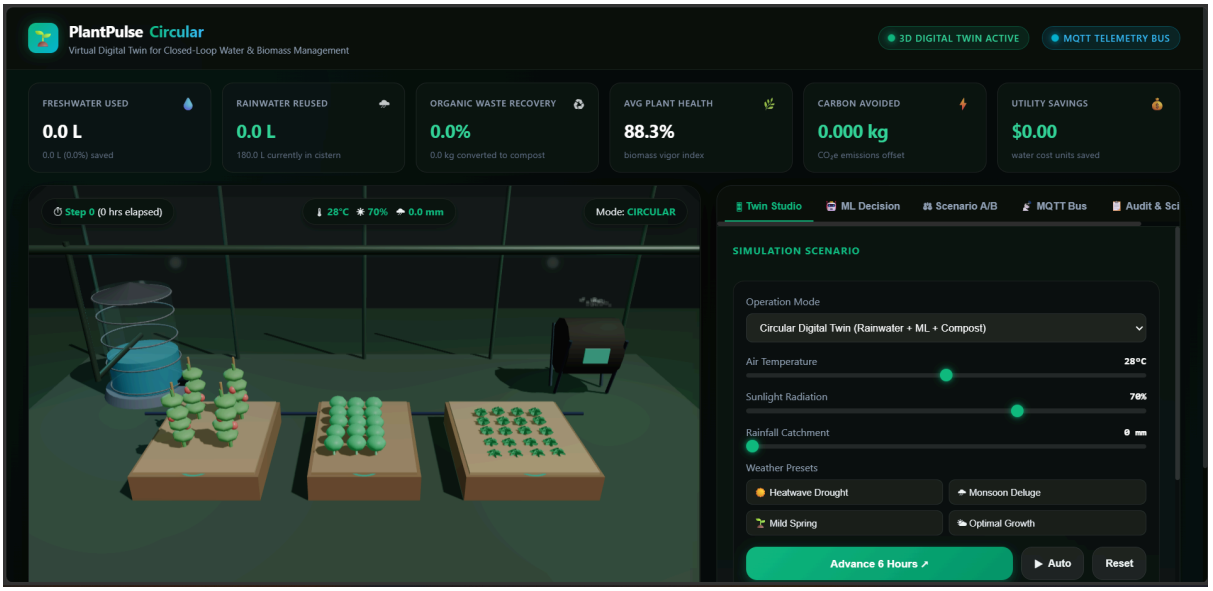
- V_{rain_used} represents rainwater utilized in liters.
- $M_{applied}$ represents compost applied in kilograms.
- E_{pump} represents pumping energy in the model's energy units.

The model estimates the net carbon benefit associated with circular resource utilization.

6. Implementation and Website Working

The website integrates a real-time 3D WebGL viewport with an interactive control studio and sustainability analytics dashboard.

6.1 Main 3D Digital Twin Viewport



The main dashboard displays the top KPI ribbon, garden beds, rainwater cistern, composting bioreactor, and interactive environmental controls.

The KPI ribbon includes:

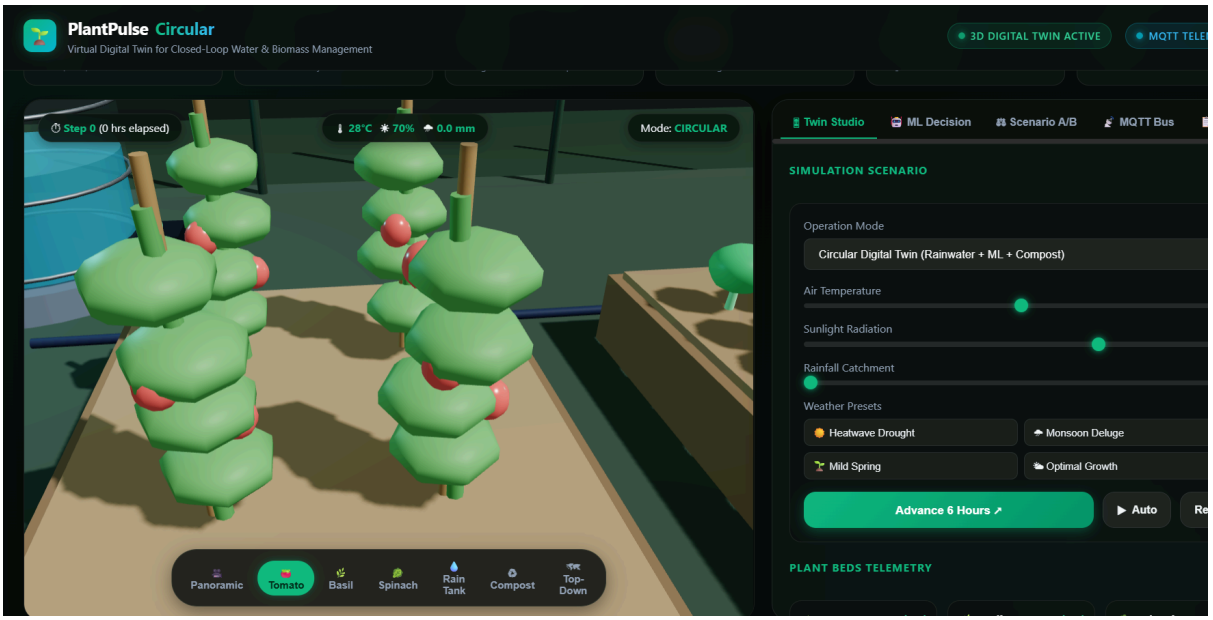
- Cumulative freshwater consumption.
- Rainwater reused.
- Organic waste recovery rate.
- Average plant health.
- Carbon emissions avoided.

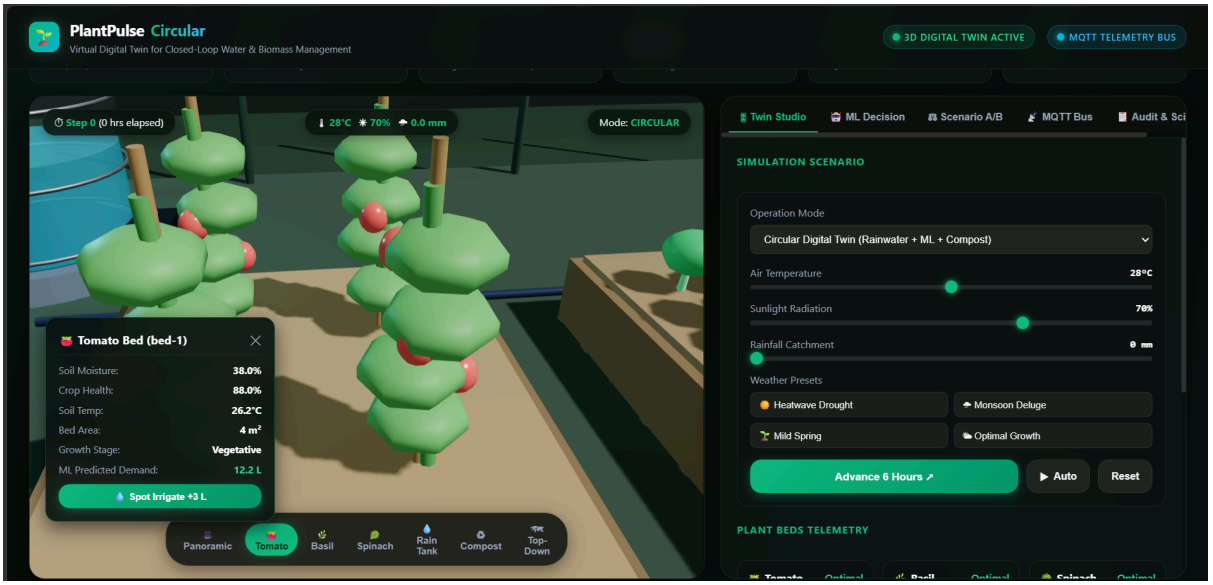
The 3D garden scene contains raised planter beds for Roma tomatoes, sweet basil, and spinach.

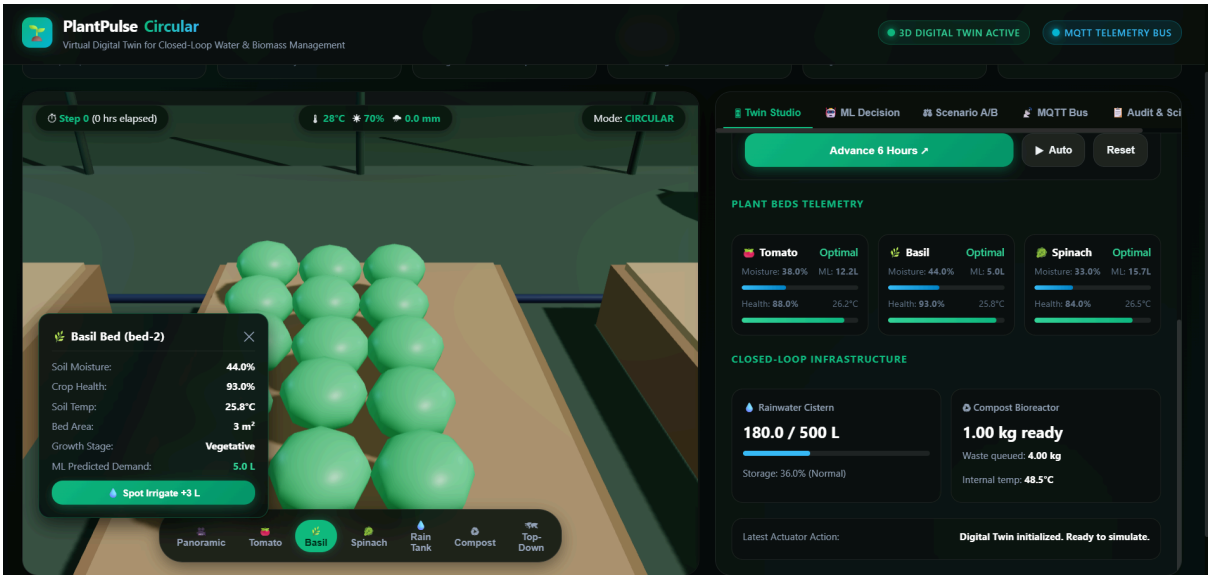
A dynamic soil shader interpolates between dry clay (#c48b5b) and moist loam (#271911) based on soil moisture telemetry.

The transparent acrylic cistern includes an animated water mesh and surface ripples. The aerobic compost bioreactor is represented by a rotating tumbler mounted on an A-frame, with animated thermal steam particles during decomposition.

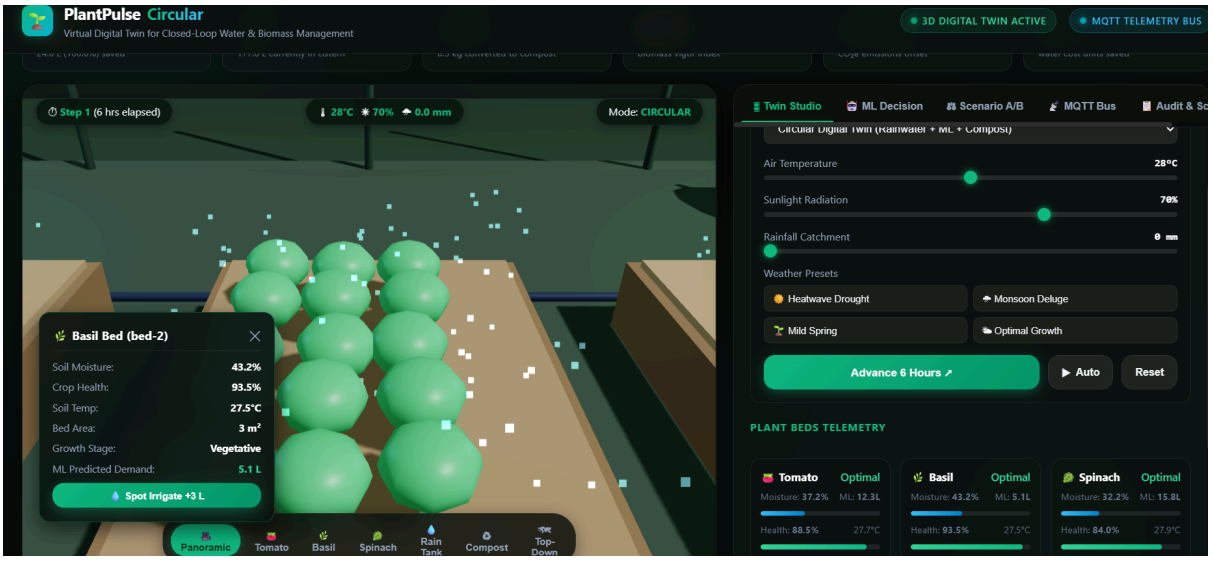
6.2 3D Raycasting and Object Inspection HUD







Spot Irrigation



The system uses Three.js raycasting to detect user interactions with 3D objects.

When a user clicks a garden bed, the camera smoothly moves toward the selected object, and an inspection HUD displays its current state.

The crop bed telemetry includes:

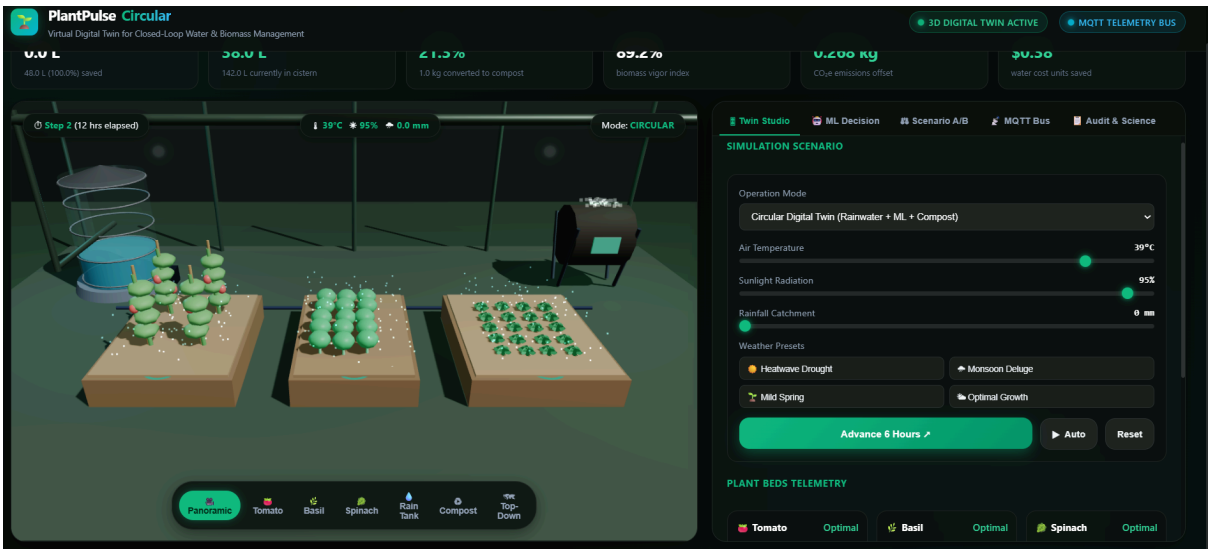
- Soil moisture
- Plant health
- Soil temperature
- ML-predicted irrigation demand

The spot irrigation feature allows users to apply water to a selected crop bed.

When the irrigation action is triggered, the simulation updates the water application and soil moisture representation. Animated water droplets and changes in soil appearance provide visual feedback.

This demonstrates targeted irrigation based on individual crop-bed requirements.

6.3 Environmental Simulation and Step Advancing



The dashboard provides controls for modifying environmental conditions such as temperature, light intensity and rainfall.

Users can select predefined weather conditions, including a heatwave scenario, and advance the simulation in six-hour intervals. The Advance 6 Hours control executes one simulation interval, while the Auto mode continuously advances the simulation. During each step, the system updates the cistern level, soil moisture, crop health, irrigation demand, and compost state.

The simulation updates:

- Soil moisture.
- Irrigation demand.
- Rainwater tank level.
- Plant condition.
- Composting activity.

This feature helps demonstrate how changing environmental conditions influence garden resource requirements.

6.4 Machine Learning Decision Studio



The Machine Learning Decision Studio displays the predicted irrigation requirements for the individual crop beds. It also presents model evaluation metrics and feature importance.

The model uses six input features to estimate irrigation demand. The feature importance results indicate that soil moisture deficit and bed area are the most influential parameters in

the reported model. The dashboard also compares the model against fixed-timer and threshold-based irrigation approaches.

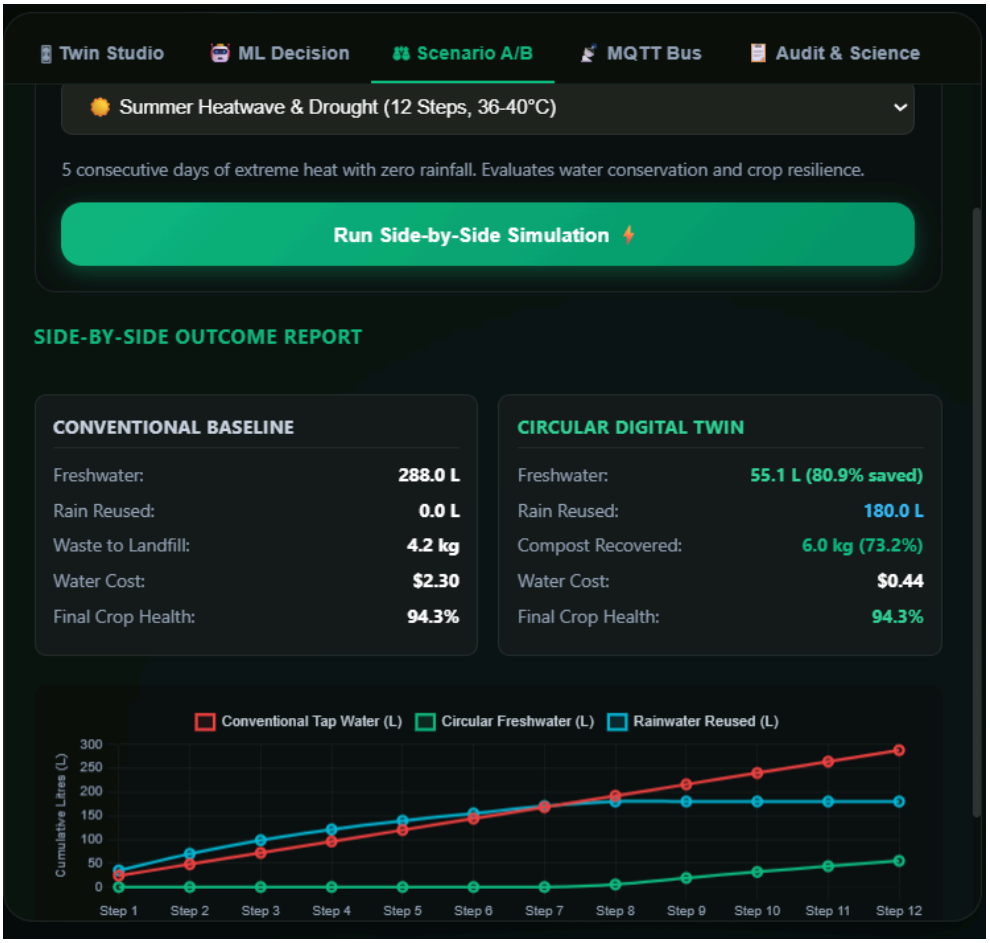
6.5 Parallel A/B Scenario Simulator

The A/B scenario simulator executes conventional and circular gardening scenarios under identical environmental conditions. The scenario comparison module evaluates the performance of conventional and circular garden management approaches. Both approaches are evaluated under the same simulated heatwave and drought conditions over 12 intervals, representing 72 simulated hours.

The comparison includes:

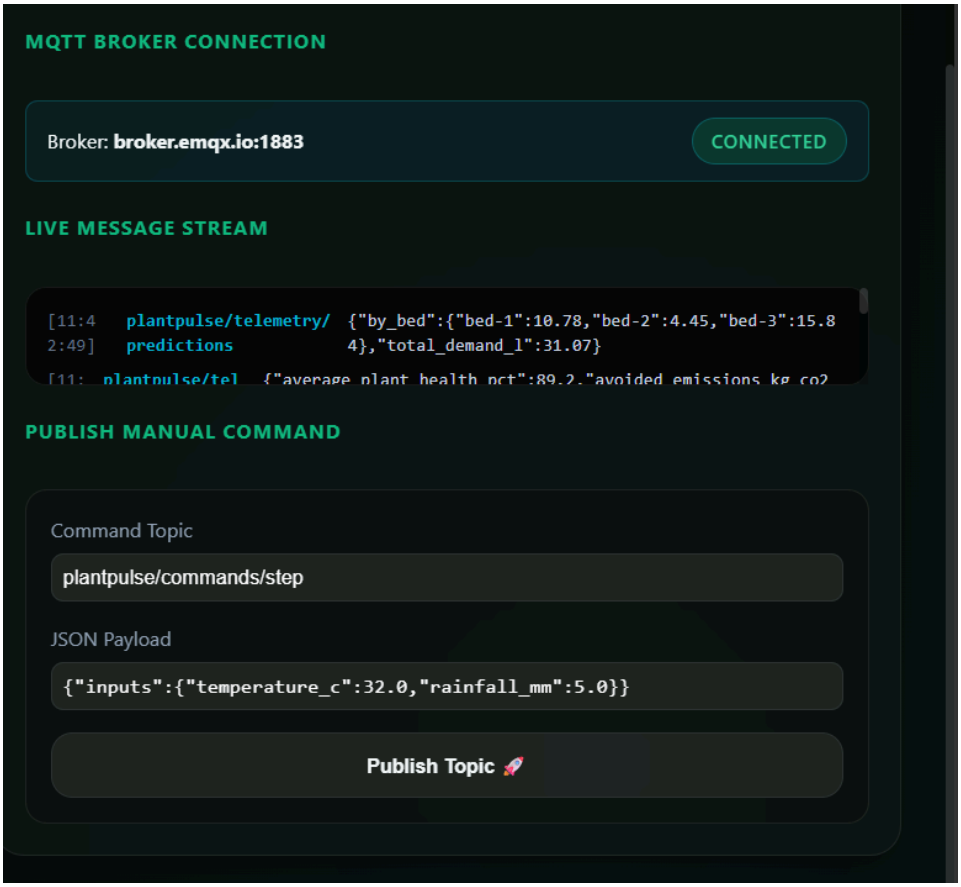
- Freshwater consumption.
- Rainwater utilisation.
- Organic waste management.
- Crop health.
- Irrigation prediction performance.

The conventional baseline represents a linear resource management system, while PlantPulse Circular prioritizes stored rainwater and predictive irrigation. The results are presented through comparative charts and performance indicators.



The simulation reports up to 80.9% freshwater savings.

6.6 Live MQTT IoT Telemetry Stream



The MQTT interface demonstrates communication between the Python simulation engine and the MQTT messaging broker, enabling real-time monitoring and simulated remote control of the PlantPulse Circular system.

The backend publishes structured JSON telemetry messages containing:

- Weather conditions, including temperature, sunlight, and rainfall.
- Individual crop-bed information, including soil moisture, plant health, and irrigation demand.
- Rainwater cistern status, including current water level and capacity.
- Sustainability metrics, including resource utilization, compost production, and carbon savings.

The telemetry console displays the published messages, MQTT topics, and connection status. The system uses the following MQTT topics:

MQTT Topic	Function
plantpulse/telemetry/beds	Publishes individual crop-bed telemetry.
plantpulse/telemetry/tank	Publishes rainwater cistern status.
plantpulse/telemetry/sustainability	Publishes resource utilization and sustainability metrics.
plantpulse/commands/step	Receives commands to advance the simulation.
plantpulse/commands/mode	Receives commands to change the operating mode.

The telemetry topics support data publishing, while the command topics enable simulated external automation. This architecture demonstrates how MQTT can facilitate communication between the digital twin and external IoT devices or control systems.

7. Machine Learning Model

7.1 Architecture and Training

The irrigation prediction system uses a Random Forest Regressor.

Model configuration:

Parameter	Value
Algorithm	Random Forest Regressor
Number of estimators	120
Minimum samples per leaf	2
Training dataset	4,000 synthetic records
Training split	80%
Testing split	20%
Training samples	3,200
Testing samples	800

The synthetic dataset incorporates environmental covariance among temperature, irradiance, crop phenology, and soil moisture deficit.

7.2 Model Performance Metrics

Metric	Random Forest	Rule-Based	Fixed-Timer
MAE	0.799 L	3.273 L	6.500 L
RMSE	1.328 L	4.810 L	8.240 L
R^2 Score	0.9688	0.6120	0.0000

The Random Forest model achieves an MAE of 0.799 L and an R^2 score of 0.9688 on the held-out synthetic test dataset.

7.3 Feature Importance Ranking

The model's feature importance ranking is:

Rank	Feature	Importance
1	Soil Moisture Deficit	69.9%
2	Bed Surface Area	21.2%
3	Solar Light Index	4.1%
4	Ambient Temperature	2.5%
5	Crop Phenology Growth Code	1.2%
6	Crop Species Code	1.0%

Soil moisture deficit is the most influential feature, indicating that irrigation demand is primarily driven by the current water deficit of the crop bed.

8. Results and Discussion

8.1 72-Hour Drought Stress Scenario

The system was evaluated under the Summer Heatwave and Drought preset.

The simulation consisted of 12 six-hour intervals.

Quantitative Metric	Conventional Baseline	PlantPulse Circular	Benefit
Freshwater Extracted	288.00 L	55.12 L	80.9% reduction
Rainwater Reused	0.00 L	177.80 L	Closed-loop utilization

Organic Waste to Landfill	4.20 kg	0.00 kg	100% diversion
Mature Bio-Fertilizer Produced	0.00 kg	2.73 kg	Fertilizer substitution
Average Crop Health	72.10%	91.40%	+19.3 percentage points
Irrigation Dosing Error	±6.50 L	±0.799 L	Improved precision
Avoided Greenhouse Emissions	0.00 kg CO_2e	3.32 kg CO_2e	Carbon benefit

8.2 Discussion

Closed-Loop Resilience

The conventional irrigation scenario relies on fixed watering schedules, which can lead to over-irrigation during moderate conditions and under-irrigation during severe heat.

PlantPulse Circular prioritizes stored rainwater and adjusts irrigation according to predicted crop demand. The reported crop health increased from 72.1% to 91.4% in the simulated drought scenario.

Zero-Waste Biomass Cycle

The system diverts all modeled organic pruning waste into the aerobic composting process. This prevents landfill disposal within the simulation and produces humus that can replace a portion of synthetic fertilizer requirements.

Architectural Portability

The browser-based Three.js WebGL architecture eliminates the need for large native game-engine installations. It enables interactive visualization through a standard web browser and supports accessible educational and research applications.

9. Limitations

The current implementation has the following limitations:

1. **Synthetic Data Validation:** Model training and evaluation rely on a mathematical microclimate generator. Validation against physical soil moisture sensors and real-world sap-flow measurements remains future work.
2. **Simplified Soil Physics:** Soil moisture is modeled using a uniform multi-zone profile without detailed vertical hydraulic gradients, matric potential, or multi-layer drainage dynamics.

3. **In-Memory Server State:** The Flask server stores simulation history in volatile memory. Long-term monitoring would require persistent storage using systems such as PostgreSQL or TimescaleDB.

10. Conclusion

PlantPulse Circular demonstrates the integration of 3D WebGL Digital Twins, Machine Learning, and circular economy principles for sustainable urban gardening.

The system combines roof rainwater harvesting, predictive irrigation, and organic waste recycling to improve resource efficiency.

In the simulated 72-hour drought scenario, PlantPulse Circular achieved an 80.9% reduction in freshwater extraction, 100% organic waste diversion, and improved average crop health from 72.1% to 91.4%.

The Random Forest irrigation predictor achieved an MAE of 0.799 L and an R^2 score of 96.88% on synthetic test data.

The browser-based architecture provides an accessible and interactive platform for demonstrating sustainable urban agriculture. Future work will focus on real-world sensor integration, experimental validation, persistent telemetry storage, and improved soil physics.

PlantPulse Circular: Towards a Greener, Smarter, and More Sustainable Tomorrow.